



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Master of Technology Branch: Computer Science and Engineering Session: 2020-21

Semester: I semester Specialization: Computer Science and Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam	Tests (Two)	Assignments/Quiz	End Sem Practical/ Viva	Practical Record/ assignment/Quiz /Presentation	
1.	MA- 104	Advanced Mathematics	3	1	-	4	70	20	10	-	-	100
2.	CS -101	Advanced Data Structures and Algorithm	3	1	-	4	70	20	10	-	-	100
3.	CS- 102	Object Oriented Technology and UML	3	1	-	4	70	20	10	-	-	100
4.	CS -103	Advanced Computer Architecture	3	1	-	4	70	20	10	-	-	100
5.	CS- 104	Advanced Computer Networking	3	1	-	4	70	20	10	-	-	100
6.	CS -105	Lab-I (CS-101,CS-103)	-	-	6	6	-	-	-	90	60	150
7.	CS -106	Lab-II (CS-102, CS-104)	-	-	6	6	-	-	-	90	60	150
		Total	15	5	12	32	350	100	50	180	120	800

L:Lecture T:Tutorial P: Practical

CS-105 : 1) Advanced Data Structures and Algorithm 2) Advanced Computer Architecture

CS-106: 1) Object Oriented Technology and UML 2) Advanced Computer Networking

Note: End Semester Examination will be of three hours



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

Postgraduate Programme

(Courses effective from Academic Year 2020-21)

SYLLABUS OF COURSES TO BE OFFERED “Master of Technology”

Disclaimer: The postgraduate programme in Master of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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MASTER OF TECHNOLOGY

I SEMESTER

COMPUTER SCIENCE AND ENGINEERING(CSE)



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BHABHA UNIVERSITY BHOPAL

MA-104

Advanced Mathematics

Unit 1 : Partial Differential Equation Solution of Partial Differential Equation (PDE) by separation of variable method, Numerical solution of PDE (Laplace, Poisson's, Parabola) using finite difference Methods.

Unit 2 : Matrices And Linear System Of Equations Solution of linear simultaneous equations by Gaussian elimination and its modification, Crout's triangularization method, Iterative methods- Jacobins method, Gauss-Seidal method, Determination of Eigen values by iteration.

Unit 5 : Calculus Of Variations Euler-Lagrange's differential equation, The Brachistochrone problems and other applications. Isoperi-metric problem, Hamilton's Principle and Lagrange's Equation, Rayleigh-Ritz method, Galerkin method.

Unit 4 : Fuzzy Logic Operations of fuzzy sets, fuzzy arithmetic & relations, fuzzy relation equations, fuzzy logics. MATLAB introduction, programming in MATLAB scripts, functions and their application.

Unit 5 : Reliability Introduction and definition of reliability, derivation of reliability functions, Failure rate, Hazard rate, mean time t future & their relations, concepts of fault tolerant analysis.

Reference Books:

1. Higher Engineering Mathematics - by Dr. B.S. Grewal; Khanna Publishers
2. Calculus of Variations - by Elsgole; Addison Wesley.
3. Applied Numerical Methods with MATLAB by Steven C Chapra, TMH.
4. Introductory Methods of Numerical Analysis by S.S. Shastry,
5. Calculus of Variations - by Galfand & Fomin; Prentice Hall.
6. Higher Engineering Mathematics by B.V. Ramana, Tata McHill.
7. Advance Engineering Mathematics by Ervin Kreszig, Wiley Eastern Edd.
8. Numerical Solution of Differential Equation by M. K. Jain
9. Numerical Mathematical Analysis By James B. Scarborough
10. Fuzzy Logic in Engineering by T.J. Ross
11. Fuzzy Sets Theory & its Applications by H.J. Zimmersoms



BHABHA UNIVERSITY BHOPAL

CS-101

Advanced Data Structures and Algorithm

Unit 1

Introduction: Basic Concepts of OOPS – Templates – Algorithm Analysis – Adt - List (Singly, Doubly and Circular) Implementation - Array, Pointer, Cursor Implementation

Unit 2

Basic Data Structures: Stacks And Queues – Adt, Implementation And Applications - Trees – General, Binary, Binary Search, Expression Search, **AVL**, Splay, B-Trees – Implementations - **Tree Traversals**.

Unit 3

Advanced Data Structures: Set – Implementation – Basic Operations On Set – Priority Queue – Implementation - Graphs – Directed Graphs – **Shortest Path Problem**- Undirected Graph - **Spanning Trees** – Graph Traversals

Unit 4

Memory Management ; Issues - Managing Equal Sized Blocks - Garbage Collection Algorithms For Equal Sized Blocks - Storage Allocation For Objects With Mixed Sizes - Buddy Systems - Storage Compaction

Unit 5

Searching, Sorting And Design Techniques: Searching Techniques, Sorting – Internal Sorting – Bubble Sort, Insertion Sort, Quick Sort, Heap Sort, Bin Sort, Radix Sort – External Sorting – Merge Sort, **Multi-Way Merge Sort**, **Polyphase Sorting** - Design Techniques - Divide And Conquer - **Dynamic Programming** - Greedy Algorithm – Backtracking - Local Search Algorithms

Reference Books :

1. Mark Allen Weiss, “Data Structures and Algorithm Analysis in C++”, PearsonP
2. Aho, Hopcroft, Ullman, “Data Structures and Algorithms”, Pearson EducationP
3. Drozdek, Data Structures and algorithm in Jawa, Cengage(Thomson)
4. Gilberg, Data structures Using C++,Cengage



BHABHA UNIVERSITY BHOPAL

CS- 102

Object Oriented Technology and UML

Unit 1

C++ preliminaries :- Tokens, Keywords, Variable, scope of variables, Data type, pointers, operators- scope resolution, member de-referencing operators, memory management operators, manipulators, typecast operators; Symbolic constants, Type compatibility, Dynamic initialization, Reference variable, Call by reference.

Unit 2

Objects & Classes:-

abstract & declaration syntax, visibility label-private, public, protected, Inline concept, Static data member & member function, Array of objects, Pointer to objects & members, Array of pointers to objects. Functions :- Declaration & definition, exploring arrays & strings, function overloading, const function, Passing & returning object through function, The Friend function.

Unit 3

Constructors & Destructors:- Default constructors, default argument constructor, parameterized constructor, Copy constructor, Destructor.

Inheritance and Polymorphism:- Visibility modes, Single Inheritance, Multi-level Inheritance, Hierarchical Inheritance, Multiple Inheritance, Hybrid Inheritance, Virtual base class, abstract class. Function Overloading, Operator overloading, overloading unary, binary, string manipulation using operators. Run time - Virtual function, pointer to object, this pointer, pure virtual function.

Unit 4

Object Modeling Technique (OMT):- object model, function model, relationship among models, object diagrams, state diagrams, data flow diagrams, Object oriented Design: Overview of object design, Combination of models, Designing algorithms, design optimization, Implementation of control, Adjustment, Design of association

Unit 5

Unified Modeling Language (UML): Class diagram sequence diagram Use case diagram, Collaboration diagram, state, chart diagram, Activity diagram, component diagram, deployment diagram, Object oriented Database: Relational Vs .object oriented database, the architecture of object oriented database, query language for Object Oriented database.



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Reference Books:

1. Rumbaugh, James Michael, Blaha - "Object Oriented Modelling and Design"- Prentice Hal India/ Pearson education.
2. Jana, C++ & Object Oriented Programming, PHI
3. OOP in C++ by Lafore, Galgotia Pub.
4. Balagurusamy; Object oriented programming with C++; TMH





BHABHA UNIVERSITY BHOPAL

CS- 103 Advanced Computer Architecture

Unit-1

Introduction: review of basic computer architecture, quantitative techniques in computer design, measuring and reporting performance.

Unit- 2

CISC and RISC processors. Pipelining: Basic concepts, instruction and arithmetic pipeline, data hazards, control hazards, and structural hazards, techniques for handling hazards. Exception handling.

Unit -3

Pipeline optimization techniques. Compiler techniques for improving performance. Hierarchical memory technology: Inclusion, Coherence and locality properties; Cache memory organizations, Techniques for reducing cache misses; Virtual memory organization, mapping and management techniques, memory replacement policies.

Unit- 4

Instruction-level parallelism: basic concepts, techniques for increasing ILP, superscalar, super-pipelined and VLIW processor architectures. Array and vector processors. Multiprocessor architecture: taxonomy of parallel architectures.

Unit- 5

Centralized shared-memory architecture: synchronization, memory consistency, interconnection networks. Distributed shared-memory architecture. Cluster computers. Non von Neumann architectures: data flow computers, reduction computer architectures, systolic architectures.

Reference Books:

1. John L. Hennessy and David A. Patterson, Computer Architecture: A Quantitative Approach, Morgan Kaufmann.
2. John Paul Shen and Mikko H. Lipasti, Modern Processor Design: Fundamentals of Superscalar Processors, Tata McGraw-Hill
3. M. J. Flynn, Computer Architecture: Pipelined and Parallel Processor Design, Narosa Publishing House
4. Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, McGraw-Hill.



BHABHA UNIVERSITY BHOPAL

CS-104

Advanced Computer Networking

Unit 1

Review of Networking and O.S. fundamentals, ISO-OSI Model, different layers and their functions, LAN, MAN, WAN, Communication media & principles IEEE standards etc.

Unit 2

Internetworking with TCP/IP, Basic concepts, Principles, Protocols and Architecture, Address handling Internet protocols and protocol layering. DNS, Applications: TELNET, RLOGN , FTP, TFTP, NFS, SMTP, POPL, IMAP, MIME, HTTP,STTP,DHCP, VOIP,SNMP.

Unit 3

Introduction to Router, Configuring a Router, Interior & Exterior Routing, RIP, Distance Vector Routing, OSPF, BGP, Uni-cast, Multicast and Broadcast. Multicast routing protocols: DVMRP, MOSPF, CBT, PIM, MBONE, EIGRP, CIDR, Multicast Trees, Comparative study of IPv6 and IPv4.

Unit 4

VPN addressing and routing, VPN Host management, ATM Concepts, Services Architecture, Equipments and Implementation

Unit 5

Introduction to wireless transmission and medium access control, wireless LAN: IEEE 802.11, HiperLAN , Bluetooth Mobile Network and Transport layer, WAP GSM and CDMA: Network architecture and management

Reference Books:

1. Computer Networks: Tanenbaum.
2. Internetworking with TCP/IP: Comer.
3. Data Communications, Computer Networks and Open Systems: Hallsall.
4. Data Communications, Stalling.
5. Mobile Communication: Schiller, Pearson Education
6. Computer Communications and network Technology, Gallo, Cengage(Thomson)
7. Wireless and Mobile Network Architecture: Yi Bing Lin, Wiley
8. ATM Network: Kasara, TMH
9. TCP/IP protocol Suite, Forouzan , CTMH



BHABHA UNIVERSITY BHOPAL

CS-105

LAB-I (CS-101, CS-103)

CS-101-→ Advanced Data Structures and Algorithm

CS-103-→ Advanced Computer Architecture

LIST OF EXPERIMENTS

- (1) Write a Program to illustrate Linked List using C++ Program.
- (2) Write a Program to illustrate stack using Linked list in C++.
- (3) Write a Program to illustrate stack using Array.
- (4) Write a Program to illustrate Bubble Sort Algorithm.
- (5) Write a Program to illustrate Binary Searching in Array.
- (6) Write a Program to illustrate Quick Sort Algorithm
- (7) Write a Program to illustrate Exception Handling

Reference Books:

- 1 Rambaugh, James Michael, Blaha - "Object Oriented Modelling and Design"- Prentice Hal India/ Pearson education.
- 2 Jana, C++ & Object Oriented Programming, PHI
- 3 OOP in C++ by Lafore, Galgotia Pub.
- 4 Balagurusamy; Object oriented programming with C++; TMH
5. John L. Hennessy and David A. Patterson, Computer Architecture: A Quantitative Approach, Morgan Kaufmann.
6. John Paul Shen and Mikko H. Lipasti, Modern Processor Design: Fundamentals of Superscalar Processors, Tata McGraw-Hill
7. M. J. Flynn, Computer Architecture: Pipelined and Parallel Processor Design, Narosa Publishing House
8. Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, McGraw-Hill.



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CS-106

LAB-II (CS-102, CS-104)

CS-102-→ Object Oriented Technology and UML

CS-104-→Advanced Computer Networking

LIST OF EXPERIMENTS

- (1) Write a Program to illustrate Object and Classes.
- (2) Write a Program to illustrate Constructor and Destructor.
- (3) Write a Program to illustrate Arrays.
- (4) Write a Program to illustrate Derived Classes & Inheritance.
- (5) To draw UML Use Case Diagram of College Information System.
- (6) To draw UML activity diagram for Library Management System.
- (7) To draw UML class diagram for ATM.
- (8) To Compare ISO-OSI Model & TCP/IP Model.
- (9) To Compare LAN, MAN, WAN.

Reference Books :

1. Mark Allen Weiss, “Data Structures and Algorithm Analysis in C++”, PearsonP
2. Aho, Hopcroft, Ullman, “Data Structures and Algorithms”, Pearson EducationP
3. Drozdek, Data Structures and algorithm in Jawa, Cengage(Thomson)
4. Gilberg, Data structures Using C++,Cengage
5. Computer Networks: Tanenbaum.
6. Internetworking with TCP/IP:Comer.
7. Data Communications, Computer Networks and Open Systems: Hallsall
8. Computer Communications and network Technology, Gallo, Cengage(Thomson)